Prob. 4. Z&T, Ch. 5, 1.1 a, b.

For each of the ↓ P.D. operators, write down the principal part, $P_m(x, D)$, & the characteristic eq-n $p_m(x, \xi) = 0$.

(a) (1.6). $P(D) = D_1^2 + D_2^2$ ← $D^{(0,2)}$

(b) (1.7). $P(D) = D_1^2 - D_2^2$

(c) (1.8) $P(D) = D_1^2 - D_2$

(d) (1.9) $P(D) = x \cdot D_1^2 + D_2^2$ ← $\boxed{D^{(0,0,2)}}$

(e) (1.11) $P(D) = D_1^2 + D_2^2 - D_3^2$

(f) $P(D) = D_2^2 + D_3^2 - D_1$

(g) $P(D) = D_1^4 + 2D_1^2 \cdot D_2^2 + D_2^4 + y \cdot D_1^2 D_2 - D_1 D_2^2$

(h) $P(D) = D_1^2 + D_2^2 + x D_3^2 - y D_1^2 D_2$.

Sol-n.

(a) Principal part: $P_m(\underline{x}, D) = D_1^2 + D_2^2$, $m = 2$ → $\begin{bmatrix} x \\ y \end{bmatrix}$

Principal symbol: $p_2(\underline{x}, \underline{\xi}) = \xi_1^2 + \xi_2^2$

Characteristic eq-n (direction) $p_2(\underline{x}, \underline{\xi}) = 0$

$$\xi_1^2 + \xi_2^2 = 0$$

(b) Pr. part $P_m(\underline{x}, D) = D_1^2 - D_2^2$, $m = 2$.

Pr. symbol $p_m(\underline{x}, \underline{\xi}) = \xi_1^2 - \xi_2^2$,

Char. eq-n. $p_m(\underline{x}, \underline{\xi}) = 0 \Rightarrow \xi_1^2 - \xi_2^2 = 0$

(c) Prin. part $P_m(\underline{x}, D) = D_1^2$, $m = 2$

Princ. symbol $p_m(\underline{x}, \underline{\xi}) = \xi_1^2$

Char. eq-n $\xi_1^2 = 0$

(d) Prin. part. $P_m(\underline{x}, D) = x_2 D_1^2 + D_2^2$ $\underline{x}$

Prin. symbol. $p_m(\underline{x}, \underline{\xi}) = x_2 \xi_1^2 + \xi_2^2$

Char. eq-n. $x \xi_1^2 + \xi_2^2 = 0$

(e) Prin. part $P_m(\underline{x}, D) = D_1^2 + D_2^2 - D_3^2$, $m = 2$

Prin. symb. $p_m(\underline{x}, \underline{\xi}) = \xi_1^2 + \xi_2^2 - \xi_3^2$

Char. eq-n $\xi_1^2 + \xi_2^2 - \xi_3^2 = 0$

(f) Prin. part $P_m(\underline{x}, D) = D_2^2 + D_3^2$, $m = 2$

Prin. Sym. $p_m(\underline{x}, \underline{\xi}) = \xi_2^2 + \xi_3^2$

Char. eq-n.

$$\xi_1^2 + \xi_2^2 - \xi_3^2 = 0.$$

(g) Prin. part. $P_m(\underline{x}, D) = D_1^4 + 2D_1^2 D_2^2 + D_2^4$, $m = 4$

Prin. symbol $p_m(\underline{x}, \underline{\xi}) = \xi_1^4 + 2\xi_1^2 \cdot \xi_2^2 + \xi_2^4$

Char. eq-n. $\xi_1^4 + 2\xi_1^2 \cdot \xi_2^2 + \xi_2^4$

(h) Prin. part $P_m(\underline{x}, D) = -y D_1^2 D_2$, $m = 3$

Prin. symb. $p_m(\underline{x}, \underline{\xi}) = -y \xi_1^2 \xi_2$

Char. eq-n. $p_m(\underline{x}, \underline{\xi}) = 0$

$$-y \xi_1^2 \xi_2 = 0.$$

$\blacksquare$